Practical 7: Professional Maps with QGIS & CRD Data

Day 4 — Duration: 3.5 Hours

SDI Training for ICT Professionals — Climate Risk Database

What you will accomplish: By the end of this practical, you will have created a professional-quality map PDF using data from the live CRD platform. You will learn to connect QGIS to CRD web services, style layers with colours and symbols, query data, create a print-ready layout with title, legend, and scale bar, and explore the CRD API in your browser.

Before you start: QGIS must be installed and working. You need an internet connection to access the live CRD at `crd.resilienceacademy.ac.tz`. If the live CRD is unavailable, you can use your local CRD at `http://localhost`.

Part 1: Connect to the Live CRD (45 minutes)

In this part, you will connect QGIS to the live Resilience Academy CRD and load real datasets.

Add a WMS Connection

Step 1: Open **QGIS Desktop** on your computer.

Step 2: Save a new project: **Project** → **Save As...** → name it `crd_map_practical.qgz` on your Desktop.

Step 3: In the menu, click **Layer** → **Add Layer** → **Add WMS/WMTS Layer...**

You should see: The "Data Source Manager — WMS/WMTS" dialog.

Step 4: Click the **"New"** button to create a new connection.

Step 5: Fill in:

- **Name:** `CRD Live`
- **URL:**

```
https://crd.resilienceacademy.ac.tz/geoserver/ows
```

What this does: Creates a connection to the GeoServer instance powering the live CRD. GeoServer serves map layers as images (WMS) that QGIS can display.

Step 6: Click **"OK"** to save.

Step 7: Make sure **"CRD Live"** is selected in the dropdown, then click **"Connect"**.

You should see: After a few seconds, a long list of available layers appears in a tree structure. These are all the datasets hosted on CRD.

Step 8: Scroll through the list and find **"Tanga_imagery_reduced"** (or search for it if a search box is available). Click on it to select it.

Step 9: Click **"Add"**.

You should see: A satellite/aerial imagery layer of Tanga appears on the QGIS map canvas. This is being streamed directly from the CRD server.

Add WFS Layers (Vector Data)

WFS (Web Feature Service) gives you the actual data (not just images), so you can query, style, and analyse it.

Step 10: In the menu, click **Layer** → **Add Layer** → **Add WFS Layer...**

You should see: The "Data Source Manager — WFS" dialog.

Step 11: Click **"New"** to create a connection.

Step 12: Fill in:

- **Name:** `CRD Live WFS`
- **URL:**

```
https://crd.resilienceacademy.ac.tz/geoserver/ows
```

Step 13: Click **"OK"**, then click **"Connect"**.

You should see: A list of vector layers available as WFS. This may take a moment to load.

Step 14: Find and select **"mwanza_businesses"** (or search for "mwanza"). Click on it.

Step 15: Click **"Add"**.

You should see: Point features appear on the map showing business locations in Mwanza. Unlike WMS, these are real vector features you can click, query, and style.

Step 16: Still in the WFS dialog, find and select **"mwanza_education_facilities"**. Click **"Add"**.

You should see: Education facility points appear on the map.

Step 17: Click **"Close"** to close the dialog.

Your Layers panel should now show:

1. mwanza_education_facilities (points)
2. mwanza_businesses (points)
3. Tanga_imagery_reduced (raster/image)

You can toggle layers on/off by clicking the checkbox next to each layer name.

Right-click on **"mwanza_businesses"** and select **"Zoom to Layer"** to centre the map on the Mwanza data.

You should see: The map zooms to show Mwanza with business and education facility points visible.

Part 2: Style the Layers (45 minutes)

Default symbols are plain and hard to read. Let us make the map look professional.

Categorised Symbols for Businesses

Step 19: In the Layers panel, double-click on the **"mwanza_businesses"** layer (or right-click → **"Properties"**).

You should see: The Layer Properties dialog with multiple tabs on the left.

Step 20: Click the **"Symbology"** tab (the paintbrush icon on the left).

You should see: The symbology settings. By default, it shows "Single Symbol" with a plain circle.

Step 21: At the top, change the dropdown from **"Single Symbol"** to **"Categorized"**.

What this does: Instead of all points having the same colour, each category (e.g., business type) will get a different colour.

Step 22: In the **"Value"** dropdown, select the column that contains the business type. This might be called `type`, `business_type`, `amenity`, or similar.

Step 23: Click the **"Classify"** button at the bottom.

You should see: A list of categories appears, each with a different colour. For example: restaurant (red), shop (blue), hotel (green), etc.

Step 24: To change colours:

- Double-click on any colour swatch next to a category
- Choose a new colour from the colour picker
- Click **"OK"**

Step 25: To change the symbol size:

- Click on the symbol preview (the small circle)
- In the symbol settings, change **"Size"** to `3.0` (millimetres)
- You can also change the shape: click the dropdown and choose a triangle, square, star, etc.

Step 26: Click **"OK"** to apply the symbology.

You should see: The map now shows businesses in different colours based on their type. The legend in the Layers panel shows the colour key.

Style Education Facilities

Step 27: Double-click the **"mwanza_education_facilities"** layer to open Properties.

Step 28: Go to **"Symbology"**.

Step 29: Keep it as **"Single Symbol"** (since these are all schools).

Step 30: Change the symbol:

- Click the symbol preview
- Change the colour to a distinct colour (e.g., dark blue or purple)
- Change the size to `3.5`
- Optionally, change the marker to a different shape (e.g., a triangle for schools)

Step 31: Click **"OK"**.

You should see: Education facilities are now clearly distinguishable from businesses on the map.

Add Labels

Step 32: Double-click the **"mwanza_businesses"** layer to open Properties.

Step 33: Click the **"Labels"** tab (the "abc" icon on the left).

Step 34: Change the dropdown from **"No Labels"** to **"Single Labels"**.

Step 35: In the **"Value"** dropdown, select the column containing the business name (e.g., `name`).

You should see: A preview of how labels will look.

Step 36: Adjust the label appearance:

- **Font:** Leave as default or choose a clean font
- **Size:** Set to `8` points
- **Colour:** Dark grey or black
- **Buffer:** Click the **"Buffer"** sub-tab, check **"Draw text buffer"**, and set size to `1.0` mm with a white colour. This adds a white outline around the text making it readable on any background.

Step 37: Click **"OK"**.

You should see: Business names appear next to each point on the map. The white buffer makes them readable even on dark backgrounds.

> **Tip:** If labels overlap and look messy, zoom in. At close zoom levels, fewer labels are visible and they are spaced better. You can also go back to Labels settings and enable **"Rendering"** → **"Merge connected lines to avoid duplicate labels"** or adjust placement options.

Part 3: Attribute Table & Queries (30 minutes)

Explore the Attribute Table

Step 38: Right-click on **"mwanza_businesses"** and click **"Open Attribute Table"**.

You should see: A spreadsheet-like table showing all the data behind the map points. Each row is one business, and columns show name, type, address, and other attributes.

Step 39: Explore the table:

- Scroll down to see all records
- Click on a column header to sort by that column
- Click on a row to highlight that feature on the map
- Note the total number of features shown at the top of the table

Filter by Attribute

Step 40: With the attribute table open, click the **"Filter"** icon (funnel icon) at the bottom-left of the table, or click the dropdown at the bottom that says "Show All Features" and change it to **"Show Features Visible on Map"**.

Step 41: To filter to a specific type, click the dropdown and select **"Field Filter"**. Select the type column, then type a value to filter by (e.g., "restaurant").

You should see: Only businesses matching your filter appear in the table.

Select Features by Expression

Step 42: Close the attribute table. In the QGIS main toolbar, click **"Select Features by Expression"** (the yellow square with an "E" and a cursor).

You should see: The "Select by Expression" dialog opens with a text area and a list of available fields and functions.

Step 43: In the expression box, type an expression to select businesses of a specific type. For example:

```
"type" = 'restaurant'
```

What this does: Selects all features where the "type" column equals "restaurant". Replace "type" and "restaurant" with the actual column name and value in your data.

Step 44: Click **"Select Features"**.

You should see: The matching features are highlighted in yellow on the map and in the attribute table. The status bar at the bottom shows how many features were selected.

Step 45: To clear the selection, click **"Deselect All"** in the toolbar (or press `Ctrl+Shift+A`).

Useful expressions:

- `"name" LIKE '%hotel%'` — Selects features with "hotel" anywhere in the name
- `"type" = 'school' AND "name" LIKE '%primary%'` — Combines two conditions
- `"capacity" > 100` — Selects features where a numeric field exceeds a value

Part 4: Create a Print Layout (1 hour)

This is where you turn your styled map into a professional document suitable for reports, presentations, or printing.

Create a New Layout

Step 45: In the QGIS menu, click **Project** → **New Print Layout...**

Step 46: When prompted for a name, type: `Mwanza Business Map`

Step 47: Click **"OK"**.

You should see: A new window opens — the Print Layout editor. It shows a blank white page (A4 size by default) with a toolbar on the left and properties panel on the right.

Add the Map

Step 48: In the left toolbar, click the **"Add Map"** button (a rectangle with a small map icon inside).

Step 49: Draw a rectangle on the page by clicking and dragging. Make it large — leave space at the top for a title and at the bottom for a legend.

You should see: Your styled map appears inside the rectangle, showing exactly what you see in the main QGIS window.

Step 50: To adjust what the map shows:

- In the right panel ("Item Properties"), find the **"Scale"** field
- Set the scale to a round number like `25000` or `50000`

- Use the **"Move Item Content"** tool (the hand icon with arrows) to pan the map within the frame

Add a Title

Step 51: Click the **"Add Label"** button in the left toolbar (the "A" icon).

Step 52: Draw a rectangle at the top of the page for the title.

Step 53: In the "Item Properties" panel on the right, find the text area and type your title:

```
Business and Education Facilities -- Mwanza City
```

Step 54: Adjust the font:

- Click **"Font"** and set the size to `20` or `24`
- Set the font to bold
- Change the colour if desired
- Set horizontal alignment to **"Centre"**

You should see: A large, centred title at the top of your map layout.

Add a Legend

Step 55: Click the **"Add Legend"** button in the left toolbar.

Step 56: Draw a rectangle in a corner of the map (or below the map) for the legend.

You should see: A legend appears showing all the symbols and their labels from your map layers.

Step 57: To clean up the legend:

- In "Item Properties", uncheck **"Auto update"** (this lets you customise which items appear)
- Select items you do not want to show (like the OpenStreetMap base layer) and click the red minus button to remove them
- Double-click on a legend item text to rename it to something more readable

Add a Scale Bar

Step 58: Click the **"Add Scale Bar"** button in the left toolbar.

Step 59: Click on the map area to place the scale bar (usually bottom-left of the map).

You should see: A scale bar showing distances in kilometres or metres.

Step 60: In "Item Properties":

- Set **"Style"** to **"Single Box"** or **"Double Box"**
- Set **"Units"** to **"Kilometres"**
- Adjust the size if needed

Add a North Arrow

Step 61: Click the **"Add North Arrow"** button (or **"Add Picture"** and select a north arrow).

Step 62: Click on the map area to place it (usually top-right of the map).

You should see: A north arrow indicating the map orientation.

Step 63: Resize it by dragging the corners. A small, subtle arrow is best.

Add a Description Text Box

Step 64: Click the **"Add Label"** button again.

Step 65: Draw a text box below the map or in the bottom margin.

Step 66: Type a description, for example:

```
This map shows the locations of businesses and education facilities in Mwanza City,
Tanzania. Data source: Climate Risk Database (crd.resilienceacademy.ac.tz). Map
created during SDI Training, March 2026.
```

Step 67: Set the font to a smaller size (e.g., `9` or `10`) and italic if desired.

Export as PDF and PNG

Step 67: To export as PDF: click **Layout** → **Export as PDF...**

Step 68: Choose a filename and location:

```
Mwanza_Business_Map.pdf
```

Save it on your Desktop.

Step 69: Click **"Save"**.

You should see: A message "Export to PDF successful". Open the PDF to see your professional map.

Step 70: To export as PNG: click **Layout** → **Export as Image...**

Step 71: Choose a filename:

```
Mwanza_Business_Map.png
```

Step 72: In the export options, set the resolution to `300` DPI for high quality.

Step 73: Click **"Save"**.

You should see: A message "Export to image successful". The PNG file is suitable for presentations and documents.

Congratulations! You have created a professional map output. Open the PDF — it should have a clear title, a well-styled map, a legend explaining the symbols, a scale bar for distance reference, and a north arrow for orientation. This is the standard expected for official reports and planning documents.

Part 5: Explore the CRD API in Your Browser (30 minutes)

The CRD has an API (Application Programming Interface) that lets you browse data programmatically. Even without programming, you can explore it in a web browser.

Browse the API

Step 73: Open your web browser and go to:

```
https://crd.resilienceacademy.ac.tz/api/v2/datasets/
```

What this does: Opens the CRD API endpoint that lists all available datasets. The API returns data in JSON format, which browsers can display in a readable way.

You should see: A JSON response (structured text in curly braces) listing datasets. Each dataset has fields like "title", "abstract", "date", "download_url", etc.

Step 74: Scroll through the results. Notice each dataset has:

- A **"title"** — the human-readable name
- A **"detail_url"** — link to more information
- A **"download_url"** — link to download the actual data
- **"keywords"** — tags describing the content

Search for Specific Datasets

Step 75: Search for Mwanza datasets by adding a search parameter to the URL:

```
https://crd.resilienceacademy.ac.tz/api/v2/datasets/?search=mwanza
```

What this does: Filters the dataset list to only show results matching "mwanza".

You should see: A shorter list showing only datasets related to Mwanza.

Step 76: Try searching for other terms:

```
https://crd.resilienceacademy.ac.tz/api/v2/datasets/?search=flood
```

```
https://crd.resilienceacademy.ac.tz/api/v2/datasets/?search=tanga
```

You should see: Different datasets matching each search term.

Find Download Links

Step 77: Click on a **"detail_url"** link for any dataset in the API results (or copy-paste it into the browser address bar).

You should see: Detailed information about that specific dataset, including download links in multiple formats (GeoJSON, Shapefile, GeoTIFF, etc.).

Step 78: Look for the **"download_url"** or **"links"** section. These URLs let you download data directly.

Step 79: Try clicking a download link to download a dataset in GeoJSON or CSV format.

Optional: Visual API Exploration with Postman

Step 80: If time permits, download and install **Postman** from `https://www.postman.com/downloads/` — it is a free visual tool for exploring APIs.

Step 81: Open Postman and create a new GET request:

```
https://crd.resilienceacademy.ac.tz/api/v2/datasets/?search=mwanza&page_size=5
```

What this does: Sends a request to the CRD API asking for 5 Mwanza-related datasets.

Step 82: Click **"Send"**.

You should see: The JSON response displayed in a formatted, collapsible tree view. Postman makes JSON much easier to read than a browser.

Step 83: Explore the response:

- Expand/collapse sections by clicking the arrows
- Note the "total" count of matching datasets
- Find the "next" link for pagination (getting the next page of results)

Why learn about APIs? APIs are how computer systems share data automatically. Understanding APIs helps you:

- Find and access data programmatically (faster than browsing the website)
- Understand how web platforms work behind the scenes
- Build automated workflows in the future (with scripts or tools like Postman)
- Connect different platforms together (e.g., KoboToolbox → CRD)

Final Checkpoint

Congratulations on completing the final practical. Confirm:

☐ **CRD connection:** QGIS connected to the live CRD via WMS and WFS

☐ **Layers loaded:** mwanza_businesses, mwanza_education_facilities, and Tanga_imagery_reduced visible in QGIS

☐ **Styled map:** Business points categorised by type with different colours; education facilities with distinct symbols; labels visible

☐ **Attribute queries:** You can open the attribute table, filter data, and select features by expression

☐ **Print layout created:** Layout includes map, title, legend, scale bar, north arrow, and description

☐ **PDF exported:** A professional map saved as `Mwanza_Business_Map.pdf`

☐ **PNG exported:** Map image saved as `Mwanza_Business_Map.png`

☐ **API explored:** You browsed the CRD API and found datasets and download links

Training Summary

Over the past four days, you have built a complete Spatial Data Infrastructure workflow:

1. **Day 1:** Set up your development environment (Docker, QGIS, terminal basics)
2. **Day 2:** Installed and customised your own CRD platform; uploaded geographic data
3. **Day 3:** Installed KoboToolbox for field data collection; built the pipeline from field surveys to CRD
4. **Day 4:** Created professional maps using CRD data in QGIS; explored the CRD API

Key skills you now have:

- Install and manage Docker containers for web platforms
- Configure and customise a GeoNode-based CRD
- Upload, document, and share geographic data through a web platform
- Design mobile data collection forms with KoboToolbox
- Transform field-collected data into publishable spatial layers
- Create professional map outputs with QGIS
- Access and explore geographic data APIs

These are the foundational skills for managing a Spatial Data Infrastructure at your institution.

End of Practicals

SDI Training for ICT Professionals — Climate Risk Database

Resilience Academy — `https://crd.resilienceacademy.ac.tz`